

History of OS & Kernel (Corrected Years)

1950s–1960s (OS Kernel se Pehle)

- No Operating System ya Kernel tha.
- Computer se interact karne ke liye:
- Punch Cards – Program likhkar machine me dalte the.
- Switches & Machine Language – Direct binary (0 aur 1) commands dete the.
- Direct hardware interaction tha – slow aur complex.

1969 – UNIX (Bell Labs)

- UNIX OS launch hua.
- Kernel introduce hua – user aur hardware ke beech mediator.
- Closed Source tha aur costly.

1981 – Microsoft (MS-DOS & Windows)

- Microsoft ne MS-DOS launch kiya.
- 1985 me Windows ka pehla GUI version aaya.
- 2001 me Windows XP popular hua.
- Windows Kernel ab bhi Closed Source hai.

1991 – Linux Kernel (Linus Torvalds)

- Linus Torvalds ne Linux Kernel banaya.
- Ye Open Source hai – public free use kar sakti hai.
- Source code GitHub par milta hai.
- Community milke secure aur reliable banati hai.

Closed Source vs Open Source

- Closed Source (UNIX, Windows) → Code available nahi.
- Open Source (Linux Kernel) → Code GitHub par available, free aur secure.

Computer Components

- CPU, RAM, HDD/SSD, Monitor, Keyboard, Mouse, Device Drivers.
- Kernel user aur hardware ke beech bridge ka kaam karta hai.

Operating System Modes

- GUI Mode → Easy, click se kaam hota hai.
- CLI Mode → Command-based, fast aur lightweight.

Performance & Usage

- Linux Kernel → Free, Secure, 24×7 available.
- Badi companies jaise Flipkart Linux par chalta hai.